

[NJC/2025/Q4]

4 A competition has three qualifying rounds. Competitors could score up to 100 points in each round. A SQLite relational database, `task4.db` has been created and populated with the data from the competition. The two tables in the database are defined as follows:

```
competitor (id, name)
scores (id*, round, score)
```

where the primary key is indicated by underling one or more attributes. Foreign key is indicated by an asterisk character `*`.

The tables have been pre-loaded with data.

Task 4.1

Write a Python program and the necessary files to to create a web application. The landing or home page of the web application should should

- display the names of all the competitors in alphabetical order.
- provide a user interface where users can enter and submit the name of the competitor whose results they are searching for
- the landing page should like this:

The image shows a web application interface with a light gray background. On the left, there is a white box titled 'Competitor List' containing a bulleted list of 20 names in alphabetical order: Ahmad Ibrahim, Desmond Tan, Evan O'Reilly, Fred Lim, Gayatri Kumar, Henry Xian, James Ling, Jin He, Joel Chang, John Smith, Kelvin Lang, Maria Phang, Mike Green, Rachel Khoo, Ravi Gopal, Regan He, Sally Jones, Siti Normah, Valerie Beh, and William Brown. On the right, there is a white box titled 'Query Results' containing a form with the label 'Competitor Name:' followed by a text input field, and a blue 'Search' button below it.

The program should perform a query on the database and render the results as follows:

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Results for Joel Chang

- Round 1: 99 points
- Round 2: 90 points
- Round 3: 65 points

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If there are no results found, display an appropriate message.

A stylesheet file named `style.css` has been created for you. You should use the styles and formatting instructions from this file in all your HTML pages. Two incomplete HTML files, `query_form.html` and `results.html`, are also provided.

Save your program code as `Task4_1_<your_name>.py` and any additional files/subfolders as needed in a folder named `Task4`.

Run the web application and save the two outputs of the program as `Task4_1a.html` and `Task4_1b.html` respectively. [17]

Task 4.2

The finalists for the competition are the top **five** competitors in round **3** of the competition. For publicity purposes, a text file needs to be created with the names and scores of the five finalists. The file should have the name and score of each finalist in the following format :

```
[{'name': 'Rachel Khoo', 'score': 91}, {...}, ... {...}]
```

Write Python code to query the database and save the results in a text file named `finalist.txt`.

Save your program code as `Task4_2_<your_name>.py` [5]